

PHYS 206 Experiment 4 Lab Report: The Photoelectric Effect Experiment

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1 Introduction and Planck's Quantum Theory

By the late 1800's many physicists thought they had explained all the main principles of the universe and discovered all the natural laws. But as scientists continued working, inconsistencies that couldn't easily be explained began showing up in some areas of study. In 1901 Planck published his law of radiation. In it he stated that an oscillator, or any similar physical system, has a discrete set of possible energy values or levels; energies between these values never occur. Planck went on to state that the emission and absorption of radiation is associated with transitions or jumps between two energy levels. The energy lost or gained by the oscillator is emitted or absorbed as a quantum of radiant energy, the magnitude of which is expressed by the equation:

$$E = hf \tag{1}$$

where E equals the radiant energy, f is the frequency of the radiation, and h is a fundamental constant of nature. The constant, h , became known as Planck's constant. Planck's constant was found to have significance beyond relating the frequency and energy of light, and became a cornerstone of the quantum mechanical view of the subatomic world. In 1918, Planck was awarded a Nobel prize for introducing the quantum theory of light.

The photoelectric effect is the emission of electrons from the surface of a metal when electromagnetic radiation (such as visible or ultraviolet light) shines on the metal. At the time of its discovery, the classical wave model for light predicted that the energy of the emitted electrons would increase as the intensity (brightness) of the light increased. It was discovered that it did not behave that way. Instead of using the wave model, treating light as a particle (photon) led to a more consistent explanation of the observed behavior.

In this lab, you will study the effect varying the light intensity and frequency has on the energy of the emitted electrons and the magnitude of the photo-current. You will also determine Planck's constant.

2 Predictions

Prior to the experiment, the following predictions were considered:

1. If the light shining on the metal becomes brighter, will the kinetic energy of the emitted electrons increase?
2. If the light shining on the metal becomes brighter, will the number of electrons being emitted per second (the photo-current) increase?
3. If the frequency of the light shining on the metal is increased, will the number of electrons being emitted per second (the photo-current) increase?
4. If the frequency of the light shining on the metal is increased, will the kinetic energy of electrons being emitted increase?
5. Will all colors (frequencies) of light cause electrons to be emitted by the metal?
6. If light with the same frequency and same intensity shines on two different metals with different work functions, will both metals emit electrons?
 - a. Will these electrons have the same kinetic energy?
 - b. Will the photo-current be larger for the metal with the smaller work function?

3 Theory

The electrons in a metal are in potential energy wells. When light shines on the metal, the energy of the photon is absorbed by the electron and converted into potential energy and kinetic energy. If the photon has sufficient energy to raise the electron out of the potential well, the electron will be ejected from the surface of the metal with kinetic energy (K) equal to the difference between the photon energy (E) and the depth of the potential well (φ , the work function of the metal):

$$K = E - \varphi \quad (2)$$

The energy of the photon is proportional to its frequency:

$$E = hf \quad (3)$$

where h is Planck's Constant.

If the energy of the photon is equal to the work function, the electron makes it out of the well with no kinetic energy and thus will not actually leave the surface of the metal. If the energy of the photon is greater than the work function, the electron makes it out of the well with some kinetic energy and the electron leaves the surface of the metal. If the energy of the photon is less than the work function, the electron does not make it out of the well.

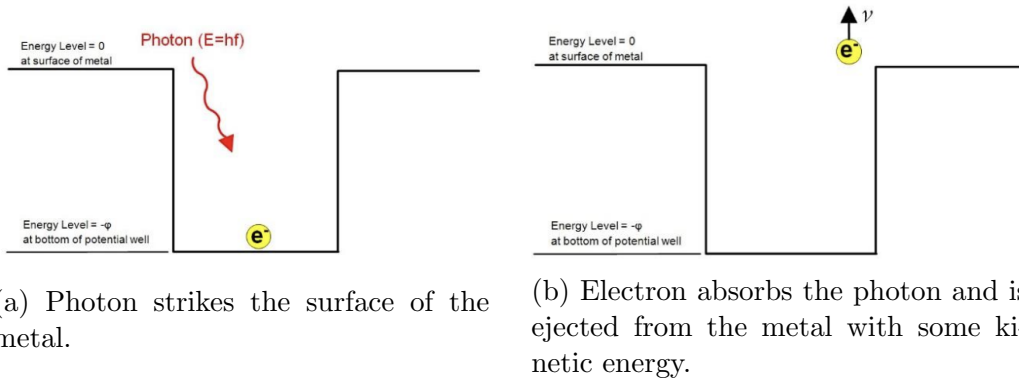


Figure 1: Photon-Electron Interaction

In this experiment, the kinetic energy of the emitted electron is measured by applying a potential difference (V) across the photodiode to just barely stop

the electron from reaching the collector plate in the photodiode tube. The voltage is adjusted until the ammeter reads zero. The kinetic energy is then equal to the potential energy, which is the charge multiplied by the potential difference:

$$K = eV \quad (4)$$

Substituting equations (3) and (4) into equation (2) gives:

$$eV = hf - \varphi \quad (5)$$

Solving for the potential difference yields:

$$V = \left(\frac{h}{e}\right) f - \frac{\varphi}{e} \quad (6)$$

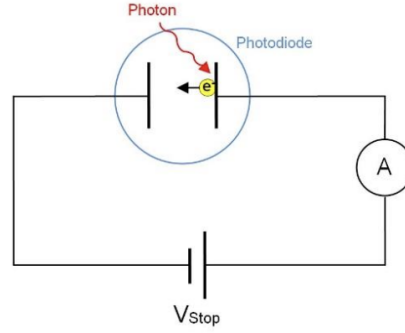


Figure 2: Photodiode Circuit

If different frequencies of light are incident on the metal, the electrons will be emitted with different kinetic energies, and thus it will take different potential differences to stop the electrons. A plot of V versus f will yield a straight line with a slope of h/e and a y-intercept of $-\varphi/e$.

Setting the voltage equal to zero and solving equation (6) for the frequency gives the x-intercept on the V versus f graph. Photons must have a frequency greater than this to cause electrons to be emitted from the metal. This is called the cutoff frequency.

$$0 = \left(\frac{h}{e}\right) f_{cutoff} - \frac{\varphi}{e} \implies f_{cutoff} = \frac{\varphi}{h} \quad (7)$$

4 Setup and Equipment

4.1 Equipment

- Mercury Lamp and Power Supply
- Photodiode
- Track
- DC Current Amplifier
- DC Power Supply I

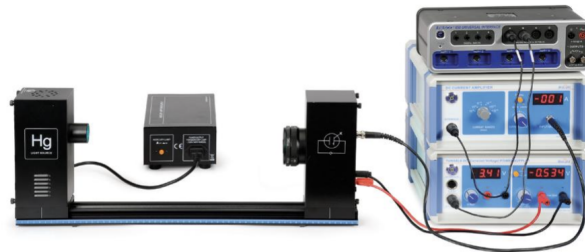
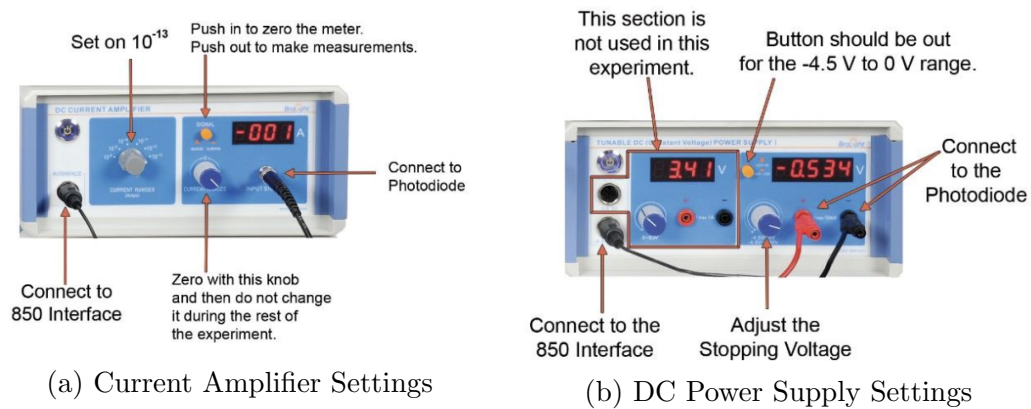


Figure 3: Experimental Setup



(a) Current Amplifier Settings

(b) DC Power Supply Settings

Figure 4: Supply Settings

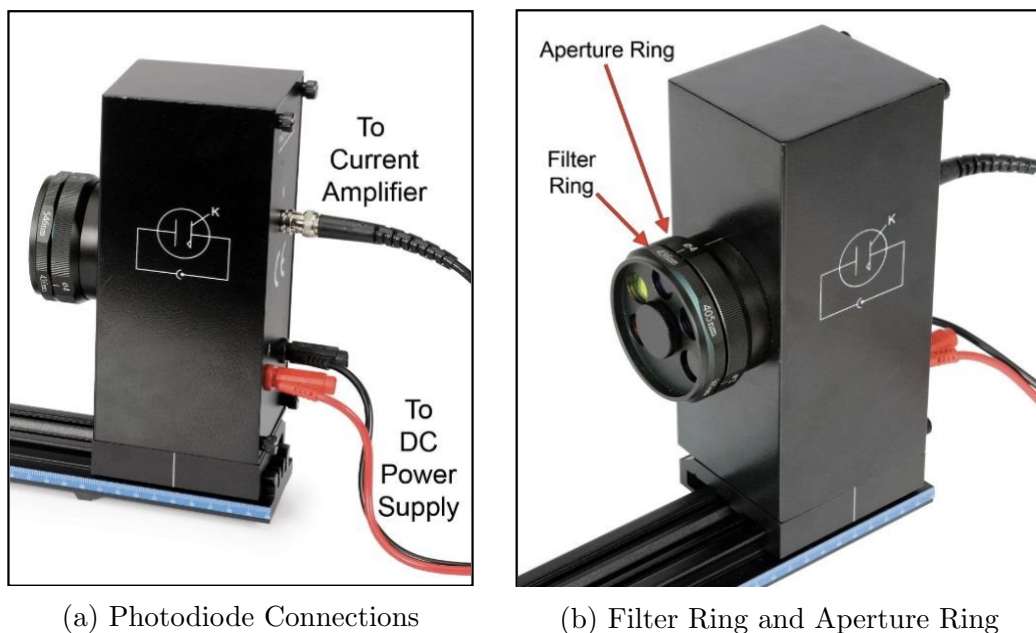


Figure 5: Photodiode Settings

4.2 Setup Procedure

1. Mount the mercury lamp and the photodiode case on the track.
2. Connect the power cord from the Mercury Light Source enclosure into the receptacle labeled "POWER OUTPUT FOR MERCURY ~220V" on the Mercury Lamp Power Supply. Connect the Mercury Lamp Power Supply to an outlet.
3. Turn on the Mercury Lamp and let it warm up for at least 10 minutes. Leave the cover on the lamp to avoid looking directly into the lamp.
4. Connect a DIN-plug-to-DIN-plug cable between Channel A on the 850 Interface and the DC Current Amplifier. Connect a DIN-plug-to-DIN-plug cable between Channel B on the 850 Interface and the DC Power Supply port labeled -4.5 V to 0 V (the lower port).
5. Do not connect any cords to the photodiode yet.
6. On the DC Current Amplifier, turn the CURRENT RANGES switch to $10^{-13}A$. Press the Calibration button in to zero the meter. Turn the

knob until the meter reads 000 A.

7. Press the Calibration button to put it in the OUT position for measuring. (NOTE: For the rest of the experiment, do not change the knobs on the DC Current Amplifier).
8. On the DC Power Supply, make sure the button is out to select the -4.5 V to 0 V range.
9. Connect the cables to the photodiode:
 - (a) Connect the special BNC-plug-to-BNC-plug cable between the port marked "K" on the Photodiode enclosure and the BNC jack on the DC Current Amplifier.
 - (b) Connect the red banana-plug patch cord between the port marked "A" on the Photodiode enclosure and the red banana jack on the right side of the DC Power Supply.
 - (c) Connect the black banana-plug patch cord between the black banana jack on the Photodiode enclosure and the black banana jack on the right side of the DC Power Supply.
10. During the experiment, you will be changing the aperture and the filters. To change the aperture, pull out on the Aperture Ring and rotate it until it clicks into place. To change the filter, do not pull out; just rotate the Filter Ring until it clicks into the next position.

5 Data Collection and Analysis

5.1 Part 1: Stopping Potential vs. Frequency

1. Select the 8 mm aperture and the 365 nm filter on the photodiode.
2. Remove the Mercury Light Source cover.
3. Adjust the VOLTAGE ADJUST knob on the DC Power Supply until the current on the ammeter is zero. Record the stopping potential.
4. Rotate the filter ring to the next longer wavelength filter and adjust the VOLTAGE ADJUST knob until the current on the ammeter is zero.

5. Repeat the procedure for all the filters and record the stopping potential values.
6. Put the cover back onto the Mercury Light Source.

Note: Frequencies are calculated using $f = c/\lambda$, where $c = 2.9979 \times 10^8$ m/s. The magnitudes of the measured negative voltages are used as the stopping potential (V_{stop}).

Table 1: Stopping Potential Measurements for Different Wavelengths

First Color	Order	Wavelength (nm)	Frequency ($\times 10^{14}$ Hz)	Stopping Potential (V) ¹
UV/Violet	1	365	8.213	-1.765
Violet	2	405	7.402	-1.313
Blue	3	436	6.876	-1.067
Green	4	546	5.491	-0.542
Yellow	5	577	5.196	-0.461

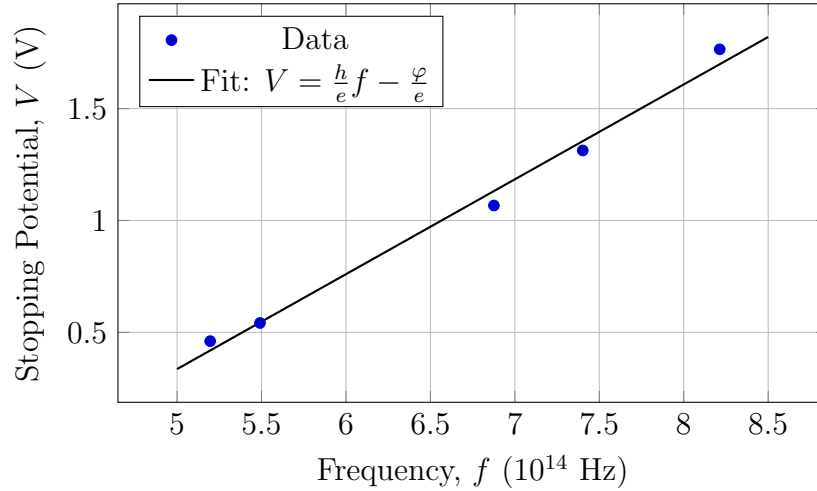


Figure 6: Linear fit of Stopping Potential versus Light Frequency.

¹The raw data reflects the negative voltage required to stop the electrons. However, for the graphical analysis and calculations below, we use the magnitude of the stopping potential, $|V_{stop}|$, since the maximum kinetic energy $K_{max} = e|V_{stop}|$ must be a positive quantity.

Calculations: From the linear fit of our data, the slope m and the y-intercept b are determined. According to equation (6), the slope corresponds to $\frac{h}{e}$.

$$h_{exp} = m \times e = (0.424 \times 10^{-14} \text{ V} \cdot \text{s}) \times (1.602 \times 10^{-19} \text{ C}) \approx 6.95 \times 10^{-34} \text{ J} \cdot \text{s}$$

The y-intercept represents $-\frac{W_0}{e}$, so:

$$W_0 = -b = 1.78397 \text{ eV}$$

5.2 Part 2: Stopping Voltage vs. Light Intensity

To investigate the effect that light intensity has on the energy of the emitted electrons, the intensity is varied by changing the aperture, while keeping the wavelength constant.

1. Enter the stopping voltage into a table for the 436 nm filter and 8 mm aperture recorded in the first procedure.
2. With the cover on the mercury light source, change the aperture from the 8 mm to 4 mm aperture on the photodiode tube.
3. Uncover the mercury light source, click on preview, and adjust the VOLTAGE ADJUST knob until the current on the ammeter is zero. Record the stopping voltage in the table.
4. Change the aperture to 2 mm and find the stopping voltage again and record it in the table.
5. Change the filter and repeat the procedure.
6. Find the average and the standard deviation of the stopping voltages.

Table 2: Stopping Potential at Varying Intensities

Color #1	%Transmission (Aperture size)	Stopping Potential (V)
Blue (436 nm)	8 mm	-1.067
	4 mm	-1.078
	2 mm	-1.095
Color #2	%Transmission (Aperture size)	Stopping Potential (V)
Green (546 nm)	8 mm	-0.541
	4 mm	-0.553
	2 mm	-0.582

6 Questions and Discussion

6.1 Part 1 Questions

1. **Which wavelength gave the emitted electrons the greatest kinetic energy? What color is this?**

Answer: The 365 nm wavelength light causes to emit electrons the greatest kinetic energy. It corresponds to the Ultraviolet/Violet part of the spectrum.

2. **From the x-intercept, find the cutoff frequency. What wavelength (color) does the cutoff frequency correspond to?**

Answer: Setting $|V| = 0$ in our linear fit equation, $f_{cutoff} = 4.20743 \times 10^{14}$ Hz. This corresponds to $\lambda = c/f_{cutoff} \approx 726$ nm, which is located in the deep red / near-infrared region.

6.2 Part 2 Questions

1. **Did the stopping voltage change as the light got brighter?**

Answer: Our experimental data show minor variation (a decrease in magnitude as light got brighter). However, it should remain constant regardless of brightness theoretically.

2. **What does the result indicate about the relationship between the kinetic energy of the electrons and the light intensity?**

Answer: The maximum kinetic energy of the emitted electrons is independent of the light intensity, contradicting the classical wave model.

3. **Describe the effect that passing different amounts of the same colored light through the Variable Transmission Filter has on the stopping potential and thus the maximum energy of the photoelectrons, as well as the charging time after pressing the discharge button.**

Answer: Passing different amounts of light only changes the number of photons hitting the metal per second. It does not affect the stopping potential, so maximum kinetic energy remains constant. However, the charging time increases as the intensity decreases because fewer electrons are emitted per second.

4. **Describe the effect that different colors of light had on the stopping potential and thus the maximum energy of the photoelectrons.**

Answer: Different colors have different frequencies. Higher frequency light resulted in a higher stopping potential, meaning the photoelectrons were ejected with a higher maximum kinetic energy.

5. **Defend whether this experiment supports a wave or a quantum model of light based on your lab results.**

Answer: This experiment supports the quantum model. The wave model predicts that higher intensity causes higher kinetic energy, which did not happen. Instead, kinetic energy increased with higher frequency, verifying the $E = hf$.

6. **Explain why there is a slight drop in the measured stopping potential as the light intensity is decreased.**

Answer: In our experimental data, we actually observed a slight increase in the stopping potential as the intensity decreased. However, the drop mentioned in the question probably would occur due to resistance and leakage currents. When the light intensity is low, the photocurrent becomes comparable to the hardware's leakage.

7 Discussion

In this experiment, we used a photodiode to measure the stopping potential of photoelectrons generated by different wavelengths of a mercury discharge lamp. By plotting the stopping potential along the frequencies, we can observe a linear relationship. The slope of this graph gave us Planck's constant experimentally, as $h \approx 6.95 \times 10^{-34} \text{ J} \cdot \text{s}$. Additionally, the y-intercept of the linear fit directly yielded the work function of the cathode as $W_0 = 1.78397 \text{ eV}$. Furthermore, testing different light intensities verified that light intensity only affects the number of ejected electrons, not their maximum kinetic energy. While our intensity data showed a minor increase in stopping potential magnitude, the overall trends strongly support the quantum model over the classical wave description. These demonstrate the quantized nature of light and the validity of $E = hf$, the fundamental principles of quantum mechanics.